

14) A company announces revised Dearness Allowance(DA) and Special Allowances (SA) for their employees as per the tariff given below:

Basic	Dearness Allowance (DA)	Special Allowances(SA)
Up to Rs.10,000	10%	5%
Rs.10,001 to Rs.20,000	12%	8%
Rs.20,001 to Rs.30,000	15%	10%
Rs.30,001 and above	20%	12%

Write a program to accept name and basic salary (BS) of an employee. Calculate and display gross salary.

Gross Salary = Basic + Dearness allowance + Special allowance

Print the information in the given format:

Name	Basic	DA	Spl. Allowance	Gross Salary
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```
import java.util.*;
class Q14{
    public static void main(){
        Scanner sc = new Scanner(System.in);
        String n;
        double basic, da, sa, gross;
        System.out.println("Enter the name");
        n = sc.nextLine();
        System.out.println("Enter the basic");
        basic = sc.nextDouble();
        if(basic <= 10000){
            da = 10.0 / 100;
            sa = 5.0 / 100;
        }else if(basic > 10001 && basic <= 20000){
            da = 12.0 / 100;
            sa = 8.0 / 100;
        }else if(basic > 20001 && basic <= 30000){
            da = 15.0 / 100;
            sa = 10.0 / 100;
        }else{
            da = 20.0 / 100;
            sa = 12.0 / 100;
        }
        gross = basic + da + sa;

        System.out.println(n + "" + basic + "" + da + "" + sa + "" + gross);
    }
}
```

15) Using a switch case statement, write a menu driven program to convert a given temperature from Fahrenheit to Celsius and vice-versa. For an incorrect choice, an appropriate message should be displayed.

Hint: $c = 5/9 * (f - 32)$ and $f = 1.8 * c + 32$

```
import java.util.*;
class Q15{
    public static void main(){
        Scanner sc = new Scanner(System.in);
        double c, f;
        int choice;
        System.out.println("MENU");
        System.out.println("1.Celsius to Fahrenheit");
        System.out.println("2.Fahrenheit to Celsius");
        System.out.println("Enter your choice 1 or 2");
        choice = sc.nextInt();
        switch(choice){
            case 1:
                System.out.println("Enter the temperature in Celsius");
                c = sc.nextDouble();
                f = 1.8 * c + 32;
                System.out.println("Temp in Fahrenheit = " + f);
                break;
            case 2:
                System.out.println("Enter the temperature in Fahrenheit");
                f = sc.nextDouble();
                c = (5 / 9.0) * (f - 32);
                System.out.println("Temp in Celsius = " + c);
                break;
            default:
                System.out.println("Invalid choice");
        }
    }
}
```

16) The volume of solids, viz cuboid, cylinder and cone can be calculated by the formula:

1. Volume of a cuboid ($v = l \cdot b \cdot h$)

2. Volume of a cylinder ($v = \pi \cdot r^2 \cdot h$)

3. Volume of a cone ($v = \frac{1}{3} \cdot \pi \cdot r^2 \cdot h$)

```
import java.util.*;
class Q16{
    public static void main(){
        Scanner sc = new Scanner(System.in);
        double l, b, h, r, v;
        int choice;
        System.out.println("MENU");
        System.out.println("1. Volume of a Cuboid");
        System.out.println("2. Volume of a Cylinder");
        System.out.println("3. Volume of a Cone");
        System.out.println("Enter your choice");
        choice = sc.nextInt();
        switch(choice){
            case 1:
                System.out.println("Enter the length");
                l = sc.nextDouble();
                System.out.println("Enter the breadth");
                b = sc.nextDouble();
                System.out.println("Enter the height");
                h = sc.nextDouble();
                v = l * b * h;
                System.out.println("Volume of a Cuboid = " + v);
                break;
            case 2:
                System.out.println("Enter the radius");
                r = sc.nextDouble();
                System.out.println("Enter the height");
                h = sc.nextDouble();
                v = (22 / 7.0) * r * r * h;
                System.out.println("Volume of a Cylinder = " + v);
                break;
            case 3:
                System.out.println("Enter the radius");
                r = sc.nextDouble();
                System.out.println("Enter the height");
                h = sc.nextDouble();
                v = (1 / 3.0) * (22 / 7.0) * r * r * h;
                System.out.println("Volume of a Cone = " + v);
                break;
            default:
                System.out.println("Invalid choice");
        }
    }
}
```

12) An employee wants to deposit certain sum of money under "Term Deposit" scheme in Syndicate Bank. The bank has provided the tariff of the scheme, which is given below:

No. of days	Rate of Interest
Upto 180 days	5.5%
181 to 364 days	7.5%
Exact 365 days	9.0%
More than 365 days	8.5%

Write a program to calculate the maturity amount taking the sum and number of days as inputs.

```
import java.util.*;
class Q12{
    public static void main(){
        Scanner sc = new Scanner(System.in);
        int amt, days;
        double rate, fin;
        System.out.println("Enter the amount");
        amt = sc.nextInt();
        System.out.println("Enter the number of days");
        days = sc.nextInt();
        if(days <= 180){
            rate = 5.5 / 100;
        } else if(days >= 181 && days <= 364){
            rate = 7.5 / 100;
        } else if(days == 365){
            rate = 9.0 / 100;
        } else{
            rate = 8.5 / 100;
        }
        fin = amt + (amt * rate);
        System.out.println("Maturity Deposit = " + fin);
    }
}
```

19) The Simple Interest (SI) and Compound Interest (CI) of a sum (P) for a given time (T) and rate (R) can be calculated as :

a) $SI = \frac{P \times r \times t}{100}$

100

b) $CI = P \times \left(\left(1 + \frac{R}{100} \right)^T - 1 \right)$

```
import java.util.*;
class Q19{
    public static void main(){
        Scanner sc = new Scanner(System.in);
        float p, r, t, si, ci;
        char choice;
        System.out.println("Enter the principal");
        p = sc.nextFloat();
        System.out.println("Enter the rate");
        r = sc.nextFloat();
        System.out.println("Enter the time");
        t = sc.nextFloat();
        System.out.println("Enter 's' for Simple Interest and 'c' for Compound Interest");
        choice = sc.next().charAt(0);

        switch(choice){
            case 's':    si = p * r * t / 100;
                        System.out.println("SI = " + si);
                        break;
            case 'c':    ci = p * (float)(Math.pow((1 + r / 100), t) - 1);
                        System.out.println("CI = " + ci);
                        break;
            default:
                System.out.println("Wrong Input");
        }
    }
}
```